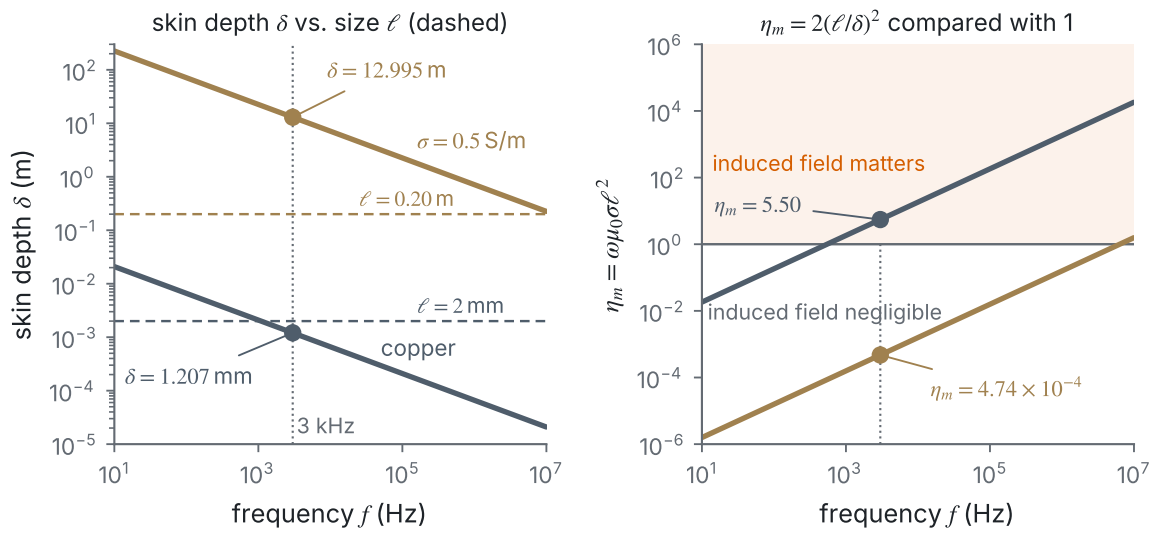
skin depth δ vs. size ℓ (dashed)
$\delta = 12.995$ m
$\sigma = 0.5$ S/m
$\ell = 0.20$ m
$\ell = 2$ mm
copper
$\delta = 1.207$ mm
3 kHz
skin depth δ (m)
frequency f (Hz)
$\eta_m = 2(\ell/\delta)^2$ compared with 1
induced field matters
$\eta_m = 5.50$
induced field negligible
$\eta_m = 4.74 \times 10^{-4}$
$\eta_m = \omega\mu_0\sigma\ell^2$
frequency f (Hz)